

RESEARCH ARTICLE

Restoration of belowground fungal communities in reclaimed landscapes of the Canadian boreal forest

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The trajectory of forests establishing on reclaimed oil sands mines in the Canadian boreal forest is uncertain. Soil microbes, namely mycorrhizal fungi, partly underlie successional trajectories of plant communities, yet their role in restoration is often overlooked. Here, we tested the relative importance of common management tools used in restoration—species planted and soil placement—on the recovery of ectomycorrhizal fungal communities over 4 years. Importantly, we further compared the community assembly of fungi on reclaimed landscapes to that in reference ecosystems disturbed to different degrees. This latter test addresses whether disturbance intensity is more important than common management interventions to restore fungal communities in these ecosystems. Three main findings emerged. (1) The effect of tree species planted and soil placement on ectomycorrhizal fungal communities establishing on reclaimed landscapes was dynamic through time. (2) Disturbances that remove or disrupt the organic layer of soils substantially affect the composition of ectomycorrhizal fungal communities. (3) Shifts in the community composition of ectomycorrhizal fungi were driven to a greater extent by disturbance severity than either tree species planted or soil placement.

Key words: community assembly, disturbance, ecological benchmarks, ectomycorrhizal fungi, forests, oil sands, reclamation

Implications for Practice

- Where possible, leave surface organic layer of soils intact to retain ectomycorrhizal fungi.
- Following restoration, expect shifts over time in ectomycorrhizal fungal community composition. Assess outcomes after at least 4 years following restoration.
- Successional trajectories of ectomycorrhizal fungal communities assembling on novel ecosystems are uncertain, thus allow for uncertainty in restoration planning.

Introduction

Novel ecosystems are those that have been created by humans and are without a historical analogue (Hobbs et al. 2009, 2014). Some landscapes in the Canadian boreal forest are arguably exemplars of novel ecosystems. Specifically, reclamation of new landscapes is occurring in areas currently mined for oil sands. Although the overall footprint is relatively small (0.02% of the boreal forest land area in Canada), disturbance by oil sands surface mining is intense. Reclamation of these mined areas includes the engineering of an entire landscape using various landforms to reconstruct physical environments upon which appropriate ecosystems are established (e.g. afforestation of uplands areas) (Johnson & Miyanishi 2008). Provincial laws regulate that following reclamation, maintenance-free, self-sustaining ecosystems with land capabilities equivalent to or higher than the pre-disturbance conditions are achieved (Government of Alberta 2017). As oil sands mining is a relatively

new disturbance in the boreal forest, successional trajectories of afforested reclaimed landscapes are uncertain (Macdonald et al. 2012, 2015; Pinno & Hawkes 2015).

Soil microbes partly underlie successional trajectories of plant communities, yet their role in restoration is often overlooked (Callaham et al. 2008; Wubs et al. 2016; Strickland et al. 2017). In particular, ectomycorrhizal (EcM) fungi are essential to ecosystem functions such as carbon and nutrient cycling as well as forest regeneration and succession (Louise & Ari 2005; Peay et al. 2008; Clemmensen et al. 2013). Ectomycorrhizal (EcM) fungi, which acquire carbon from and enhance the nutrient uptake of their hosts (Peay et al. 2008), are recognized as playing a critical role in seedling survival (Hawkins et al. 2015), nutrition (Jones et al. 2010), and mediating the fertility, structure, and turnover of organic matter in soils (Johnson et al. 2017). Thus, EcM fungi can potentially facilitate revegetation and the development of soils, two key goals in restoration. A number of factors are likely to affect the composition

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of EcM fungal communities including the presence of particular host species (Ishida et al. 2007; Huang et al. 2014; Teder-soo et al. 2016), variation in soil characteristics (Berg & Smalla 2009; Dickie et al. 2009; Ding et al. 2011), and time since disturbance (Dickie et al. 2013; Barnes et al. 2016; Kalucka & Jagodzinski 2016); these factors are also imitated in restoration. Specifically, restoration plans often include decisions on revegetation (i.e. species planted/seeded), soil management (i.e. salvage, placement, and use of amendments), and time. Time may be viewed as a timeline required to achieve the restoration goals, but also as “do-nothing and wait” strategy, which can be effective in some cases (Jones et al. 2018). Thus, these common management tools used in restoration—species planted, soil placement, and time—are likely to influence the composition of EcM fungal communities, with consequences for successional trajectories of plant communities.

Despite evidence for the beneficial roles mycorrhizal fungi play in plant establishment and performance (Brundrett et al. 1996), land reclamation efforts are often hampered by challenging site conditions (Danielson & Visser 1989), and the response of EcM fungi in disturbed sites has been mixed (Lewis et al. 2005; Jones 2017). Soils used in the restoration of surface mines can greatly differ from mature intact forests, where reclaimed soils show lower moisture, pH, and nutrient availability (Bus-sler et al. 1984; Bois et al. 2005) as well as a depleted fungal propagule bank (Dimitriu et al. 2010; Onwuchekwa et al. 2014). While some studies reported a positive response of EcM fungi following restoration of surface mines (e.g. increase in inoculum potential, shift toward late stage EcM fungi with age of reclaimed sites, increase in tree seedling biomass) (Daniel-son & Visser 1989; Kalucka & Jagodzinski 2016; Kumar & Atri 2018), other studies reported the opposite (e.g. lower EcM fungal richness, lower EcM colonization rates, decreased tree seedling survival) (Danielson 1991; Bois et al. 2005; Gebhardt et al. 2007; Onwuchekwa et al. 2014). And yet other studies reported a neutral effect of soils used in the restoration of mine sites on the dominant EcM fungal community (Huang et al. 2012, 2014, 2018). As such, it is critical to determine the extent to which species planted, soil placement, and time interact to shape the community structure of EcM fungi. Consequently, successful recovery of the ecosystem will depend on the resistance of EcM fungi to mining disturbance as well as the rate in which these EcM fungal communities re-establish to their initial composition.

Here, we test the relative importance of common management tools—selection of species planted, soil placement, and time—used in restoration on the recovery of EcM fungal communities in reclaimed landscapes. We further compare the community assembly of EcM fungi on reclaimed landscapes to that of reference forest ecosystems disturbed to different degrees associated with forest harvesting, a relatively common disturbance in the boreal forest. Aboveground disturbance severity over otherwise intact soils ranged from (1) trees and forest floor intact, (2) trees removed with forest floor intact, to (3) trees and forest floor removed. This latter test addresses whether disturbance severity is more important than common management

interventions to restore belowground fungal communities of boreal forests in post-mined landscapes.

Methods

Overview of Study

We planted 1-year-old tree seedlings of three host species at two sites: a “reclaimed” and “reference” site to assay soils for EcM fungi over a 4-year period. Both sites were large scale, that is, at least 36 ha in size to parallel the spatial scales represented in reclamation operations. We used the reclamation site to investigate the influence of time, cover soil placement, and host species on EcM fungal community composition. The reference site included several ecological benchmarks ranging in disturbance severity to which we compared the establishment of fungal communities on the reclaimed site.

Study Region

The study is located in the Central Mixedwood subregion of the Canadian boreal forest, characterized by long harsh winters (average $<-12^{\circ}\text{C}$) and short warm summers (average $\sim 10^{\circ}\text{C}$) (Beckingham et al. 1996). Specifically, over the years of our study, growing season (June–August) temperatures ranged from 6 to 34°C , with total growing season rainfall of 219 mm (data collected by O’Kane Consultants). Forests occur in uplands whereas bogs and fens occur in lowlands and depressions. Upland forests consist primarily of *Picea glauca* (Moench) Voss, *Pinus banksiana* Lamb., and *Populus tremuloides* Michx. with understories mainly comprised of *Cornus sericea* L. and *Rosa acicularis* Lindl., while bogs are dominated by *Picea mariana* Mill. Britton, Sterns, & Pogenb., and fens by both *P. mariana* and *Larix laricina* (Du Roi) K. Koch. Upland soils are classified as Luvisolic or Bunisolic and typically contain a thin eluvial A horizon, distinct Bm or Bt horizons, and an underlying C horizon. Lowland soils are poorly to very poorly drained and dominated by organic matter (peat) (Beckingham et al. 1996).

Reclamation Site

The study area was 36 ha in size and was located within the Aurora North mine ($57^{\circ}19'20''\text{N}$, $111^{\circ}30'24''\text{W}$) on the Syncrude Canada Ltd. lease in the Athabasca oil sands region of Alberta (approximately 75 km north of Fort McMurray, Alberta, Canada) (Fig. S1). The study was designed to investigate the broader impacts of various practices involving different cover soil configurations for their use in reclamation of “lean oil-sands” (i.e. bitumen $<7\%$ hydrocarbons) overburden. As part of the current study, three cover soils were tested: forest floor material (FFM, a combination of salvaged litter layer and the A and B horizons to a depth of 15 cm), and a sandy sub-soil material (B and C horizons combined to a depth of 100 cm), both salvaged from *P. banksiana*-dominated upland forests with Brunisol soils. The third cover soil was peat salvaged from organic soils to a depth of 200 cm in *P. mariana*-dominated

Table 1. Summary of soil characteristics (mean $\pm$ SE) for the reclamation and reference sites. Soils were measured across three cover soil treatments for the reclamation site constructed within the Aurora North mine on the Syncrude Canada Ltd. Lease in the Athabasca oil sands region of Alberta, Canada. The reference site was a mature *Pinus banksiana* forest disturbed to different degrees associated with forest harvesting approximately 5 km from the reclamation site. A BDL indicates measure was below detectible limits. Soil sampling was conducted by NorthWind Land Resources, Inc. in early summer 2012 and summarized under Project #11-381 (Syncrude Canada Ltd. Aurora North Capping Study) in 2013.

	Organic Matter (%)	Total Organic Carbon (%)	Nitrogen (%)	C:N ratio	Phosphorus (mg/kg)	pH	Texture
Reclamation site							
FFM ($n = 9$)	2.6 $\pm$ 0.9	1.3 $\pm$ 0.1	0.1 $\pm$ 0.1	30.2 $\pm$ 0.5	24.6 $\pm$ 0.5	5.6 $\pm$ 0.1	Sand
Peat ($n = 9$)	34.1 $\pm$ 1.6	17.0 $\pm$ 0.8	0.8 $\pm$ 0.1	25.4 $\pm$ 5.0	5.0 $\pm$ 0.1	7.4 $\pm$ 0.1	Sand
Subsoil ($n = 9$)	BDL	BDL	BDL	BDL	10.9 $\pm$ 0.5	7.2 $\pm$ 0.1	Sand
Reference site							
Trees, forest floor, and soils intact ($n = 9$)	16.5 $\pm$ 5.4	9.6 $\pm$ 3.2	0.2 $\pm$ 0.1	27.7 $\pm$ 5.9	34.2 $\pm$ 3.2	5.5 $\pm$ 0.1	Sand
Trees removed ($n = 9$)	11.1 $\pm$ 4.1	5.6 $\pm$ 2.1	0.2 $\pm$ 0.1	19.0 $\pm$ 4.6	32.1 $\pm$ 6.0	5.7 $\pm$ 0.1	Sand
Trees and forest floor removed ($n = 9$)	0.2 $\pm$ 0.1	0.1 $\pm$ 0.1	0.2 $\pm$ 0.1	4.3 $\pm$ 0.5	52.3 $\pm$ 1.9	5.9 $\pm$ 0.1	Sand

lowland forests. All three cover soils were directly placed in 1-ha plots onto the site during the winter prior to tree planting. Each 1-ha plot was replicated three times for each cover soil and contained three 25 m $\times$ 25 m tree plots planted with a single-species of either *P. banksiana*, *P. glauca*, or *P. tremuloides* (see Hankin et al. 2015; Scott et al. 2019 and Table 1 for more detailed information on cover soil and reclamation site characteristics and prescription).

To investigate the relative importance of time, host identity, and cover soils in structuring the community composition of EcM fungi present on the site, we planted three host species in each of the three cover soils and harvested them over 4 years. Specifically, single-species tree plots were planted in May 2012 with 1-year-old nursery-grown container stock of *P. tremuloides* (6 cm plug diameter, 15 cm depth), *P. glauca* (6 cm plug diameter, 15 cm depth), or *P. banksiana* (4 cm plug diameter, 12 cm depth) at a 1 m spacing equivalent to a density of 10,000 stems per hectare. These seedlings were initially colonized by two species of ectomycorrhizal fungi, *Amphinema byssoides* and *Thelephora terrestris*, most likely established in the nursery (Hankin et al. 2015). Seedlings were grown at Smoky Lake Forest Nursery (Smoky Lake, Alberta, Canada) from a mixture of open-pollinated seed collected from populations in the Fort McMurray region (Alberta, Canada). Subsets of seedlings were harvested in August 2012, 2013, and 2015. Whole seedlings were excavated, roots separated from shoots at their base, and following lateral roots outward, approximately 300 fine roots were collected. In August 2012, a total of 27 seedlings were randomly sampled (three tree species $\times$ three cover soils $\times$ one sample per plot $\times$ three plots), whereas a total of 135 seedlings (three species $\times$ three cover soils $\times$ five samples per plot $\times$ three plots) were randomly harvested during the end of each growing season in August 2013 and 2015. All data were pooled by plot.

Reference Site

To provide context for the shifts in EcM fungal community composition observed at the reclamation site, we planted 1-year-old seedlings of the same species and stock in soils of ecological benchmarks representing a range of disturbance severity. Our goal here was to evaluate the magnitude and extent

of shifts in fungal communities at the reclamation site compared with those across a range of disturbances associated with forest harvesting. Importantly, this comparison allows us to determine the relative importance of ecological factors—host species, cover soils, and disturbance severity—underlying community establishment of EcM fungi.

The reference site was a mature (~63 years old) *P. banksiana* forest overlying Eutric Brunisol soils and consisting of an *Arctostaphylos uva-ursi* and *Vaccinium* sp. understory, located approximately 5 km from the reclamation site (57°21'49"N, 111°25'45"W) (Fig. S2). The reference site consisted of three treatments ranging in aboveground disturbance severity over otherwise intact soils: (1) trees and forest floor intact, (2) clearcut harvest with trees removed but forest floor intact, and (3) clearcut harvest with trees and forest floor removed. The clearcut area had been harvested approximately 17 years prior to this study. Uniform plots in sizes equivalent to the surface soil plots on the reclamation site were selected, treatments were replicated three times, separated by a minimum of 20 m, and each contained three 2.5 m $\times$ 2.5 m plots. Similar to the reclamation site, in May 2012, each plot was planted with six to eight 1-year-old nursery-grown container stock of *P. tremuloides*, *P. glauca*, and *P. banksiana* using the same seedling stock as at the reclamation site. Seedlings at the reference site were harvested in a similar manner to ones at the reclamation site, where whole seedlings were excavated, roots separated from shoots at their base, and following lateral roots outward, approximately 300 fine roots were collected. A total of 223 seedlings (three species $\times$ three treatments $\times$ one to six samples per plot $\times$ three plots) were sampled in August 2015. All data were pooled by plot.

For all harvested seedlings at both sites and across all years, roots were separated from shoots and kept on ice until arrival at the University of Alberta where roots were stored at -20°C until further processing.

Molecular Identification of Ectomycorrhizal Fungi

All roots were thawed at 4°C and gently rinsed under tap water over a 1.2-mm sieve to remove adhering soil. Cleaned roots were cut into approximately 1-cm fragments, placed in water-filled

petri dishes (10 cm × 10 cm × 5 cm) and homogenized. Root fragments were examined under a dissecting microscope followed by further examination under a compound microscope at 100× magnification for the presence of EcM fungi on fine root tips according to color, tip shape, branching pattern, texture, and variation in hyphal and mantle structure (Goodman et al. 1998). Approximately, 300 root tips were sampled per seedling and a total of two to five EcM root tips per seedling were collected for DNA extraction (see Appendix S1 for details on selecting root tips).

The identity of EcM fungi colonizing roots was determined through Sanger sequencing. Across years, molecular methods used to sequence DNA were kept relatively constant with minor deviations in protocols (see Appendix S1 for detailed description and Table S1 for detailed description of primers used). Briefly, for each seedling, genomic DNA from each of 2–5 fresh root tips was extracted using Sigma Extraction Solution and Neutralization Solution B following manufacturer protocols (Sigma Aldrich, St. Louis, MO, U.S.A.). A nested polymerase chain reaction (PCR) amplification of the internal transcribed spacer (ITS) region of fungal nuclear rDNA was performed in 16-μL reactions using primers NSA3-NLC2 and NSI1-NLB4 (Martin & Rygiewicz 2005) for samples harvested in August 2012 at 10-μM concentration each, 1.0 μL of DNA extract, 5.4 μL of sterile MilliQ water, and 8 μL of Sigma Red Taq (Sigma Aldrich). Thermal cycling conditions for the first amplification (NSA3-NLC2) were as follows: initial denaturation at 95°C for 5 minutes, 30 cycles of denaturation (95°C for 1 minute 30 seconds), annealing (67°C for 1 minute), and extension (72°C for 1 minute 30 seconds), followed by a final extension step of 72°C for 10 minutes. Second amplifications (NSI1-NLB4) were performed with an initial denaturation at 95°C for 5 minutes, followed by 27 cycles of denaturation (95°C for 1 minute 30 seconds), annealing (55°C at 1 minute) and extension (72°C at 1 minute 30 seconds), followed by a final extension step of 72°C for 10 minutes. For samples harvested in August 2013 and 2015, amplification of the ITS region of fungal nuclear rDNA was performed using primers ITS1-F and ITS4 (Gardes & Bruns 1993) with similar cycling conditions. All amplified products were verified on either a 1 or 1.7% agarose gel and were cleaned enzymatically using ExoSAP-IT (USB/Affymetrix Inc., Santa Clara, CA, U.S.A.). Cycle sequencing was performed using BigDye Terminator v3.1 (Applied Biosystems, Foster City, CA, U.S.A.) with either NSI1-NLB4 or ITS1-F-ITS4 primers. Sequencing reactions were cleaned using ethylenediamine tetraacetic acid/ethyl alcohol and run on an ABI Prism 3,730 Genetic analyzer (Applied Biosystems).

Bioinformatic Analysis

Initial sequence editing, quality filtering, sequence clustering, and taxonomic identities of sequences were processed using the QIIME pipeline v1.9 (Caporaso et al. 2010). In brief, sequences were converted from .ab1 to .fasta and .qual file format using Geneious v10 (Biomatters Ltd., Auckland, New Zealand). Sequences were edited using the *add_qiime_labels.py*

to modify the sample ID for all .fasta sequences and ensuring uniqueness in .fasta labels. Files were preprocessed using the *convert_fastaqual_fastq.py* and paired-end sequences per dataset were merged using *join_paired_ends.py* with a minimum overlap of 10 bp. Joined sequences were processed separately per dataset using *split_libraries_fastq.py* with a minimum Phred quality score of 25. Merged sequences per dataset were clustered into operational taxonomic units (OTUs) using *pick_open_reference_otus.py* (Rideout et al. 2014) using a 97% similarity threshold with the *suppress_taxonomy_assignment* flag. Any resulting singleton OTUs were included as these were representative of fungal species based on previous morphotyping of root tips. Representative sequences per dataset were combined using the *merge_otu_tables.py*, and taxonomic affiliations were assigned by searching representative sequences from each OTU against GenBank and UNITE + INSD databases using the BLAST option in *assign_taxonomy.py*. OTUs were cross-referenced by morphotype and verified as ectomycorrhizal using the FunGuild database (Nguyen et al. 2016). Sequences of all EcM fungal OTUs were submitted in the GenBank database under accession numbers (KJ938030-KJ938040; MG754913-MG754958) (see Table S2 for a list of OTUs).

As using different primer sets across sampling years has the potential to bias analyzing the effects of time on changes in EcM fungal community structure, coverage of ITS primer sets was evaluated based on in silico PCR using ecoPCR v1.0.0 (Boyer et al. 2016) against sequences downloaded from the NCBI database. Sequences downloaded contained only full-length fungal 5.8S, 18S, and 28S sequences. Duplicate sequences were removed using QIIME v1.9 for a total of (1) 46,546 rDNA sequences tested against NSA3/NLC2 and NSI1/NLB4, (2) 35,590 rDNA sequences tested against ITS1-F/ITS4 and NSI1/NLB4, and (3) 56,303 rDNA sequences tested against ITS1-F/ITS4. Specifically, three mismatches to each ITS primer set were tolerated but sequences containing consecutive five or more “N” were excluded. Results of in silico PCR were evaluated independently at the phylum, order, family, genus, and species taxonomic levels. Coverage of PCR primer sets was calculated in terms of the proportion of fungal taxa amplified in silico compared to sequences of all EcM fungal OTUs in the current study. All primer sets showed 100% coverage across phylum, order, family, and genus levels (Fig. S3). Likewise, ITS1-F/ITS4 showed 100% coverage at the species level, whereas primer sets (1) NSA3/NLC2 and NSI1/NLB4 and (2) ITS1-F/ITS4 and NSI1/NLB4 showed slightly less coverage (88%) at the species level. However, sequences not amplified to the species level by either of the primer sets constituted only 2% of the total EcM fungal community in our study. Overall, in silico PCR analysis indicated that the coverage of ITS primer sets used was high and the taxonomic bias of amplification is expected to be low.

Statistical Analysis

All statistical analyses were carried out using R v3.3.2 (R Development Core Team 2018). To test the effects of time,

host identity, and cover soils on EcM fungal OTU richness, a generalized linear mixed model with a Poisson distribution was performed using the *glmer()* function in the *lme4* R package with a χ^2 test (Bates et al. 2015). The number of root tips was set as a random effect to account for uneven root tip distributions among plots. For the reference site, a similar model was used to test the effects of time, host identity, and cover soils on EcM fungal OTU richness. To determine the relative importance of time, host identity, and cover soils in shaping the community structure of EcM fungi at the reclamation site, a permutational multivariate analysis of variance was performed based on a presence/absence species matrix using the *adonis()* function in the R package *vegan* (Oksanen et al. 2013) with Bray–Curtis distances and permutations set to 9,999. Permutations were constrained by the number of root tips within plots using the “*strata*” argument within the *adonis()* function to account for possible effects related to uneven root tip distributions among plots. For the reference site, a similar model was used to determine the influence of disturbance severity and host identity in structuring the community composition of root-associated EcM fungi (2015 sampling period only). Disturbance severity consisted of four treatments: (1) trees and forest floor intact, (2) trees removed with forest floor intact, (3) trees and forest floor removed, and (4) reconstructed soils. The fourth treatment consisted of the three cover soil treatments from the reclaimed site, pooled by plot. The host identity treatment at the reclamation site was pooled by plot. To visualize differences in EcM fungal communities across treatments, we used non-metric multidimensional scaling ordination based on Bray–Curtis dissimilarity using the *metaMDS()* function in the *vegan* package (Oksanen et al. 2013). Variance partitioning analysis was conducted to estimate the relative contribution of time, host identity, and cover soils in shaping EcM fungal composition using the *varpart* function in the R package *vegan* (Oksanen et al. 2013). A similar model was also used to determine the relative contribution of disturbance severity and host identity in shaping EcM fungal composition at the reference site. Disturbance severity consisted of the four treatments aforementioned.

Indicator species analysis was performed to identify EcM fungal OTUs with pronounced responses to time (number of years post-planting), host species, cover soils, and disturbance severity using the *multipatt* function in the R package *indispices* (Cáceres & Legendre 2009) based on a presence/absence species matrix and permutations set to 9,999. To determine if particular groups of taxa were gained or lost as a function of disturbance severity, we calculated a ratio reflecting the number of OTUs gained or lost for a particular order by dividing the total number of gains by the total number of losses. Values greater than 1 indicated gains exceeded losses and values less than 1 indicated losses exceeded gains. Nonparametric *t* tests with 9,999 Monte Carlo permutations were used to statistically evaluate the gain or loss of fungal orders present on the roots of seedlings from mature forest with soils intact ($n = 9$) at the reference site compared to stands disturbed to different degrees associated with forest harvesting at the reference site and mining ($n = 9$) at the reclamation site.

Table 2. Number of OTUs of ectomycorrhizal fungi from roots of 1-, 2- and 4-year-old *Populus tremuloides*, *Picea glauca*, and *Pinus banksiana* seedlings planted across three reclamation cover soils (FFM, peat, and subsoil) and across a range of disturbances (trees, forest floor, and soils intact; trees removed; trees and forest floor removed). An N/A indicates measure not available.

	# of OTUs (Mean $\pm$ SE)			
	Reclamation Site	Reference Site		
		Trees, Forest Floor, and Soil Intact	Trees Removed	Trees and Forest Floor Removed
Years post-planting				
One	2 $\pm$ 1	N/A	N/A	N/A
Two	2 $\pm$ 1	N/A	N/A	N/A
Four	4 $\pm$ 1	7 $\pm$ 1	6 $\pm$ 1	4 $\pm$ 1
Host species				
<i>P. banksiana</i>	3 $\pm$ 1	8 $\pm$ 1	7 $\pm$ 2	4 $\pm$ 1
<i>P. glauca</i>	3 $\pm$ 1	7 $\pm$ 1	7 $\pm$ 2	3 $\pm$ 2
<i>P. tremuloides</i>	3 $\pm$ 1	4 $\pm$ 1	4 $\pm$ 1	3 $\pm$ 2
Soil type				
FFM	3 $\pm$ 1	N/A	N/A	N/A
Peat	3 $\pm$ 1	N/A	N/A	N/A
Subsoil	3 $\pm$ 1	N/A	N/A	N/A

Results

Influence of Time, Host Identity, and Cover Soils on Ectomycorrhizal Fungal Communities at the Reclamation Site

Overall, the EcM fungal community consisted of 35 OTUs across the reclaimed site (Table S2). Total EcM fungal OTU richness was 3 $\pm$ 1 (mean $\pm$ SE) per plot. Ectomycorrhizal fungal OTU richness varied across host seedlings differing in age ($\chi^2 = 12.38$; $p = 0.002$), with EcM fungal OTU richness highest on 4-year-old seedlings compared to 1- and 2-year-old seedlings (Table 2). Similarly, EcM fungal community composition varied across time ($F = 2.59$; $p = 0.001$) (Table S3; Fig. 1). Ectomycorrhizal fungal composition on 1-year-old seedlings differed from those on 2- ($p < 0.001$) and 4-year-old seedlings ($p < 0.001$), while EcM composition on 2-year-old seedlings differed from those on 4-year-old seedlings ($p < 0.001$). In particular, *Rhizopogon rubescens* and *Hebeloma hiemale* were EcM fungal OTU indicator species of 1-year-old seedlings, while *Hebeloma ingratum* was the sole indicator of 2-year-old seedlings (Table 3). Compared to 1- and 2-year-old seedlings, 4-year-old seedlings hosted eight EcM fungal OTU indicators including *Amphinema byssoides* and *Suillus* spp.

EcM fungal OTU richness was similar across the three host species ($\chi^2 = 0.27$; $p = 0.86$) and cover soils ($\chi^2 = 0.33$; $p = 70.84$) (Table 2). In contrast, EcM fungal community composition differed among the three host species (Fig. 1). In particular, *Cenococcum geophilum*, *H. ingratum*, and *Sphaerospora brunnea* were indicators of *Populus tremuloides* seedlings, while *P. banksiana* primarily hosted *Thelephora terrestris*, *R. rubescens*, and *Suillus luteus* (Table 3). In contrast, *A. byssoides* was the only indicator of *Picea glauca* seedlings. Furthermore, for a given host species, EcM

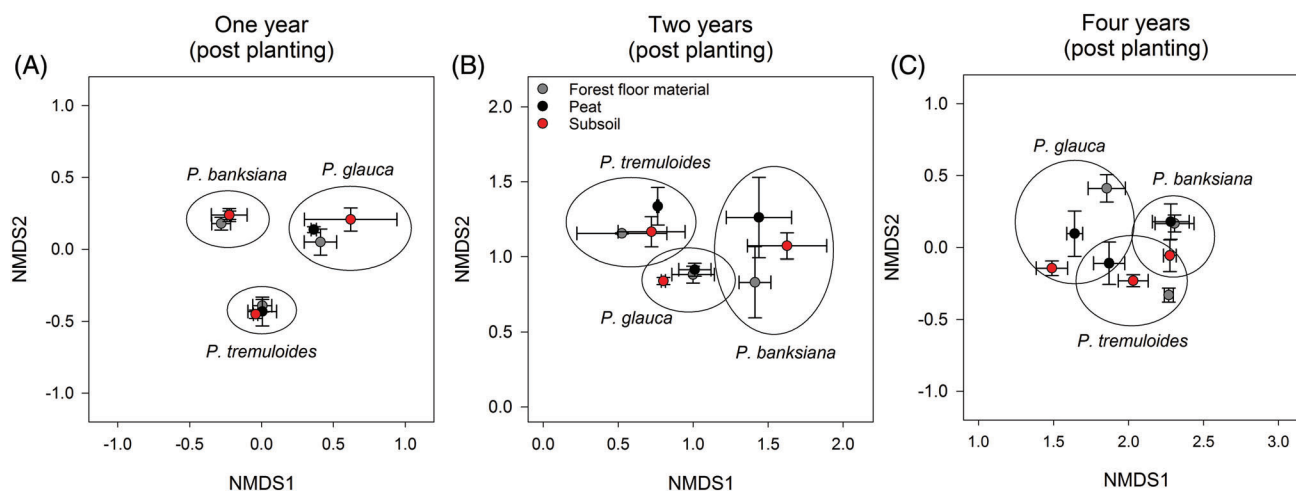


Figure 1. Non-metric multidimensional scaling ordinations visualizing shifts in ectomycorrhizal fungal communities on (A) 1-, (B) 2-, and (C) 4-year-old seedlings of *Populus tremuloides*, *Pinus banksiana*, and *Picea glauca* across three reclaimed cover soil treatments (FFM [gray centroids], peat [black centroids], and subsoil [red centroids]). Error bars indicate $\pm$ SE of each centroid.

Table 3. Indicator species analysis of OTUs of ectomycorrhizal fungi from roots of 1-, 2- and 4-year-old *Populus tremuloides*, *Picea glauca*, and *Pinus banksiana* seedlings planted across three reclamation cover soils (FFM, peat, and subsoil) and across a range of disturbances (trees, forest floor, and soils intact; trees removed; trees and forest floor removed; and reconstructed soils). Reconstructed soils consisted of the three cover soil treatments from the reclaimed site, pooled by plot. ⁺An indicator value of 1 indicates a species found in all samples of one group and no samples in any other group. *p* values were calculated based on a Monte Carlo significance test of observed maximum indicator values for each OTU. Blank cells represent an OTU was not an indicator for a specific treatment type. Data for host identity consisted of both the reference and reclamation sites. Only the 2015 sampling period (4-year-old seedlings) was used to identify strongly responding ectomycorrhizal fungal OTUs to disturbance severity and host identity.

OTU	Years Post-Planting	Host Species	Soil Type	Disturbance Severity	Indicator Value ⁺	<i>p</i>
<i>Cenococcum geophilum</i>		<i>P. tremuloides</i>	FFM		0.67	0.001
<i>Thelephora terrestris</i>		<i>P. banksiana</i>	Peat		0.60	0.002
<i>Rhizopogon rubescens</i>	One	<i>P. banksiana</i>			0.64	0.001
<i>Hebeloma hiemale</i>	One	<i>P. tremuloides</i>			0.58	0.001
<i>Hebeloma ingratum</i>	Two				0.39	0.03
<i>Sphaerospora brunnea</i>		<i>P. tremuloides</i>			0.47	0.004
<i>Amphinema byssoides</i>	Four	<i>P. glauca</i>			0.63	0.001
<i>Suillus luteus</i>	Four	<i>P. banksiana</i>			0.49	0.007
<i>Piloderma olivaceum</i>	Four		FFM		0.47	0.004
<i>Tuber</i> sp. 1	Four		Peat	Reconstructed soils	0.43	0.01
<i>Tomentella</i> sp. 2	Four				0.43	0.01
<i>Hebeloma incarnatum</i>	Four		Subsoil		0.41	0.03
<i>Russula emetica</i>	Four				0.39	0.04
<i>Suillus</i> sp. 1	Four				0.39	0.04
<i>Piloderma lanatum</i>				Trees, forest floor, and soils intact	0.64	0.001
<i>Tylospora</i> sp. 1				Trees, forest floor, and soils intact	0.64	0.001
<i>Lactarius scrobiculatus</i>				Trees, forest floor, and soils intact	0.58	0.01
<i>Piloderma bicolor</i>				Trees removed	0.69	0.002
<i>Tomentella bryophila</i>				Trees removed	0.59	0.01
<i>Hygrophorus agathosmus</i>				Trees and forest floor removed	0.48	0.04

fungal composition did not differ among cover soils on 1- and 2-year-old seedlings. However, the effect of cover soil changed with time as EcM fungal community composition differed among FFM, peat, and subsoil on 4-year-old seedlings (Fig. 1). In particular, 4-year-old seedlings in FFM primarily hosted *Piloderma olivaceum*, while 4-year-old seedlings in peat and subsoil primarily hosted *Tuber* sp.1 and *Hebeloma incarnatum* (Table 3). Variance partitioning analyses revealed that the independent, but not shared, effects of host identity

(14%), time (10%), and cover soils (3%) accounted for less than half of the explained variation in EcM fungal community composition at the reclaimed site (Fig. 2).

Comparing Ectomycorrhizal Fungal Communities at Reclaimed and Reference Sites

Overall, the EcM fungal community consisted of 52 OTUs (Table S2) across the reference site, with total EcM fungal OTU richness being 6 ± 1 (mean $\pm$ SE) per plot. Ectomycorrhizal

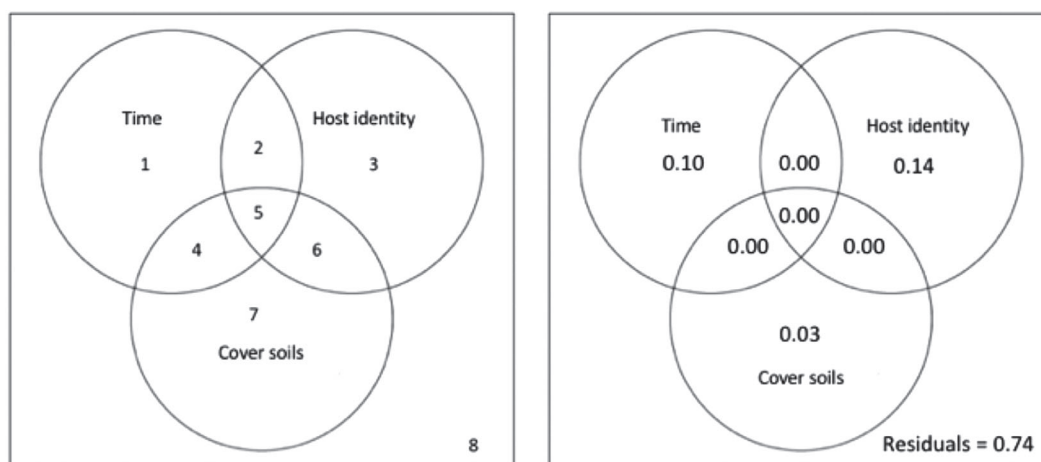


Figure 2. Variance partitioning of the effects of time, host identity, and reclaimed soils on the community structure of ectomycorrhizal fungi. Values show the percentage of variation explained by (1) time, (2) shared contribution of time and host identity, (3) host identity, (4) shared contribution of time and reclaimed soils, (5) shared contribution of time, host identity, and reclaimed soils, (6) shared contribution of host identity and reclaimed soils, (7) reclaimed soils, and (8) the residual unexplained variation.

fungal OTU richness varied by disturbance severity ($\chi^2 = 24.20$; $p < 0.0001$), with EcM fungal OTU richness being highest, where trees and forest floor were intact or where trees were removed but the forest floor was left intact (Table 2). In contrast, EcM fungal OTU richness was lowest where trees and forest floor had been removed or where trees, forest floor, and soils had been removed (i.e. reclaimed sites). However, EcM fungal OTU richness was similar across the reference site for the three host species ($\chi^2 = 9.36$; $p = 0.15$).

Ectomycorrhizal fungal community composition varied by disturbance severity ($F = 2.97$; $p = 0.001$) and host species ($F = 1.91$; $p = 0.004$) (Table S4; Fig. 3). In particular, *Piloderma lanatum*, *Tylospora* sp. 1, and *Lactarius scrobiculatus* were all indicators of mature, intact forest (Table 3). Whereas *Piloderma bicolor* and *Tomentella bryophila* were indicators of forest where trees were removed, *Hygrophorus agathosmus* was an indicator of forest where trees and forest floor were removed. Surprisingly, only *Tuber* sp. 1 was found as an indicator of reconstructed soils (i.e. reclaimed site). Furthermore, EcM fungi on seedlings planted in soils where trees were removed but forest floor and soils left intact did not differ in composition from seedlings planted into soils of mature intact forests ($p = 0.147$) (Fig. 3). The independent effects of disturbance severity (18%) accounted for greater total variation in EcM fungal community composition than host identity (7%) (Fig. 4). That is, although we observed some variation in EcM fungal community composition among the tree species to assay soils, disturbance severity accounted for most of the explained variation. In particular, there was a greater loss of fungal species belonging to Atheliales (33–100%), Agaricales (33–67%), and Russulales (50–75%) with trees, forest floor, and soils removed than those belonging to Boletales and Myrtiliniidiales (Table 4). In contrast, species belonging to Pezizales, Sebaciniales, and Thelephorales increased in frequency with severity of disturbance. These latter species were most prevalent at the reclamation site.

Discussion

We tested the relative importance of common management tools used in restoration—species planted, soil placement, and time—on the recovery of EcM fungal communities on a reclaimed oil sands mine. We subsequently compared the community assembly of EcM fungi on a novel reclaimed landscape to that of a reference forest ecosystem disturbed to different degrees. Three main findings emerged. (1) The effect of tree species and soil placement on EcM fungal communities establishing on reclaimed landscapes was dynamic through time. (2) Disturbances that removed or disrupted the organic layer of soils substantially affect the composition of EcM fungal communities. (3) Shifts in the community composition of EcM fungi were driven to a greater extent by disturbance severity than tree species planted or soil placement. Taken together, our results indicate that common management interventions in restoration cannot alone reestablish EcM fungal communities of target ecosystems on reclaimed sites.

Dynamic Effects Through Time of Tree Species and Soil Placement on Ectomycorrhizal Fungal Community Composition

At the reclaimed site, host identity influenced the community composition of EcM fungi in the first 2 years following planting, whereas the cover soil used in soil placement became more important 4 years following planting. Many EcM fungi have broad host ranges, while others only associate with specific host species (Molina & Trappe 1982). Different tree species can provide distinct niches for fungal symbionts and have been found to play a significant role in structuring the community composition of mycorrhizal fungi (Morris et al. 2009; Martínez-García et al. 2015). For example, EcM fungi have shown preference for different tree species in wet sclerophyll forests of Tasmania (66%) (Tedersoo et al. 2008), conifer–broadleaf forests in Japan (15%) (Ishida et al. 2007), neotropical rainforests in Ecuador (19%)

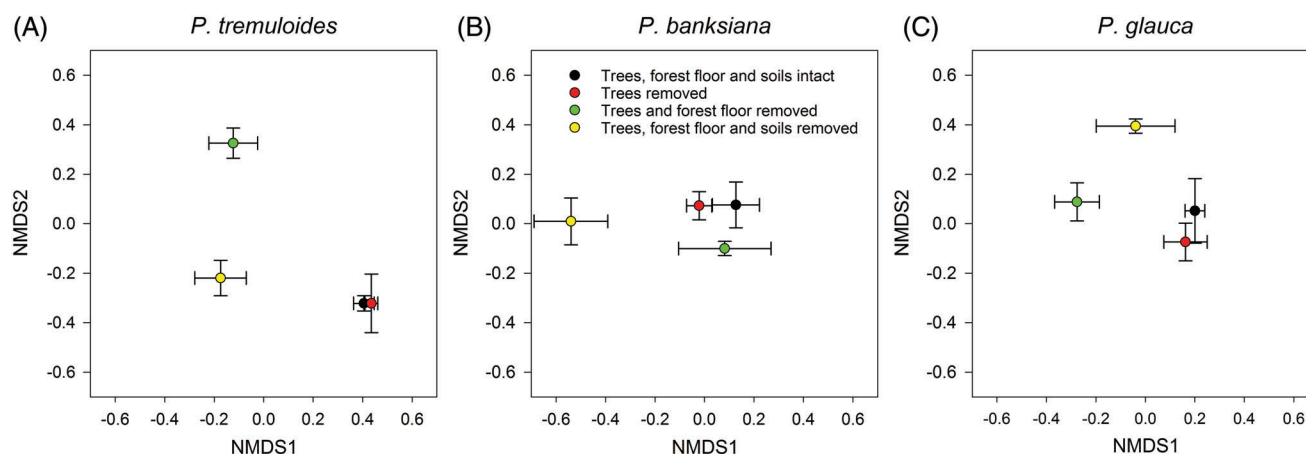


Figure 3. Non-metric multidimensional scaling ordinations visualizing shifts in ectomycorrhizal fungal communities on (A) *Populus tremuloides*, (B) *Pinus banksiana*, and (C) *Picea glauca* outplanted seedlings across a range of disturbances (trees, forest floor, and soils intact [black centroids], trees removed [red centroids], trees and forest floor removed [green centroids] and reconstructed soils [yellow centroids]). Error bars indicate $\pm$ SE of each centroid.



Figure 4. Variance partitioning of the effects of disturbance severity and host identity on the community structure of ectomycorrhizal fungi. Values show the percentage of variation explained by (1) disturbance severity, (2) shared contribution of disturbance severity and host identity, (3) host identity, and (4) the residual unexplained variation.

(Tedersoo et al. 2010), and subtropical evergreen broadleaved forests of southwest China (33%) (Ding et al. 2011). However, in our study, 2 years following tree planting at the reclaimed site, fungal communities developed quickly and were subsequently influenced by the type of cover soil in which seedlings were planted. These results suggest that EcM fungal inoculum was similar across soils initially and fungi colonizing roots were sorted by host identity. This initial pool of fungi was likely those species able to withstand soil handling such as displacement, mixing, and transport. Interestingly, it seems those fungi able to withstand severe disturbance are host-generalists, such as *Amphinema*, *Suillus*, and *Tomentella* spp. With time, we observed increases in fungal richness perhaps reflecting delayed germination of fungal propagules in salvaged soils or dispersal limitation from the regional pool. Importantly, cover soil type seemed to influence which fungi persisted following these processes, dormancy, and dispersal. However, although the influence of cover soil on EcM fungal community composition became more pronounced with time, the strength of host

identity was consistent across the study duration and explained a considerably greater portion of variance than the type of cover soil used in reclamation.

Disrupted Organic Layer of Soils Substantially Affect the Composition of Ectomycorrhizal Fungal Communities

Our results suggest that communities of EcM fungi in natural boreal jack pine forests are resistant following the removal of trees with retention of the forest floor compared with treatments that remove trees and disrupt the organic layer of soils. This result is surprising given that EcM fungi depend on living trees as a carbon source. The primary conclusion from a review of the impacts of clearcut logging on EcM fungi was that it changes the species composition of the fungal community (Jones et al. 2003; Jones 2017). However, geographic and edaphic factors can outweigh the effects of harvesting on microbial communities (Kranabetter et al. 2017; Wilhelm et al. 2017). For instance, 11–17 years following harvest, the majority of

Table 4. OTUs of fungal orders gained and lost following different degrees of disturbance severity. Gain/loss ratio reflects the number of OTUs gained or lost for a particular order by dividing the total number of gains by the total number of losses. Values greater than one indicate that gains exceeded losses and values less than one indicate that losses exceeded gains. Nonparametric *t* tests with 9,999 Monte Carlo permutations were used to statistically evaluate the gain and loss of fungal orders present on seedlings from mature forest with soils intact ($n = 9$) compared to stands where trees were removed, trees and forest floor were removed, or reconstructed soils where trees, forest floor, and soils were removed and salvaged soils placed back ($n = 9$). * $p < 0.05$.

Order	Gain/Loss Ratio		
	Trees Removed	Trees and Forest Floor Removed	Reconstructed Soils
Total EcM fungi	1.11	1.04	0.85*
Agaricales	1.00	0.75*	0.67*
Atheliales	0.33*	0.67*	0.00*
Boletales	2.00*	2.00*	1.50*
Myrtiliniales	1.00	2.00*	1.00
Pezizales	0.50*	0.50*	2.00*
Russulales	1.75*	0.25*	0.50*
Sebacinales	1.00	2.00*	2.00*
Thelephorales	1.25*	1.00	1.50*

microbial taxa were unaffected by timber harvesting across 18 reforested long-term soil productivity study sites (Wilhelm et al. 2017). Only when whole trees were removed and the upper organic layer of forest floor scraped away were fungal communities differentiated from those of reference forests. However, most of the variation in fungal community composition in this broad-scale, long-term study was explained by geography (Wilhelm et al. 2017). Considering the scale and range of treatments applied in our study, our findings are aligned with those above. Specifically, scraping away the forest floor is not a typical forest management practice and its effects would not be captured by studies on clearcuts. As our study was located within a single region in the boreal forest, any effects of biogeography would be removed. The lack of difference in the composition of EcM fungi in mature forest with soils intact versus trees removed with soils intact will need to be tracked over time for further clarity.

Disturbance Severity as a Driver of Ectomycorrhizal Fungal Community Composition

The effects of disturbance we observed on mycorrhizal colonization of plants agree with previous research examining soils disturbed by mining (Gardner & Malajczuk 1988; Jasper et al. 1991; Malajczuk et al. 1994). That is, shifts in the community composition of EcM fungi were driven to a greater extent by disturbance severity than either host identity or cover soil used in soil placement. Although planting trees of different species on the reclaimed sites recovered slightly different EcM fungal communities across the reconstructed soils, there remained a wide gap between communities assayed in novel reclaimed ecosystems and those of ecological benchmarks representing less soil disturbance. This suggests that reclamation likely has

the most pronounced influence on EcM fungal communities, while manipulations of cover soil type and tree species planted played a relatively minor role. In consequence, the need for nursery inoculation of seedling with effective EcM fungi for tree plantings into reclaimed areas (Malajczuk et al. 1994; Cordell et al. 2002) and warrants further investigation. Of course, policies regulate that reclamation occurs along with soil placement and revegetation (Government of Alberta 2017). Beyond these coarse restoration practices and given the increase in fungal richness across the study duration, it may be that time is a promising tool to restore fungal communities in these reclaimed landscapes. Specifically, the recovery of EcM fungal communities on reclaimed sites may increase with time since disturbance.

Coupled with monitoring biotic richness and composition, we suggest that additional management objectives may be warranted when restoring novel ecosystems. Specifically, a shift from focus on pre-disturbance system characteristics to one of ecosystem function should be considered. For instance, across a range of disturbances (wildfire, clearcut, and intact forests), functional complementarity among EcM fungi resulted in similarity in overall activity of hydrolytic enzymes involved in releasing absorbable forms of carbon, nitrogen and phosphorus from soil organic matter (Jones et al. 2010). Despite shifts in the composition of EcM fungal communities across these disturbances, EcM function was maintained. Likewise, soil enzyme activity was similar across pine forests of North America, regardless of the rapid community turnover of fungi at local scales (Talbot et al. 2014). This functional convergence among EcM fungi may provide a high degree of resilience in the face of rapid ecosystem change. Functional redundancy is a particularly common community structure–function relationship observed in mycobiomes seemingly built on the high α - and β -diversity of soil fungi (Peay et al. 2016). Looking forward, we suggest goals for restoring belowground fungal communities establishing in novel ecosystems include function.

While disturbance severity and host identity together explained 25% of the variation in EcM fungal structure, 75% of the variation was unexplained and likely demonstrates that there were other unmeasured factors such as: (1) soil abiotic properties (Tedersoo et al. 2011, see Table 1), (2) seedling quality, growth, and development (Merlin et al. 2019), and/or (3) colonizing vegetation (e.g. graminoids) (Bockstette et al. 2018; Merlin et al. 2019) which might have acted independently or interacted with changes in disturbance severity and/or host identity. Furthermore, in our study, the reclamation and reference sites were isolated in that the distance to undisturbed forests exceeded 1 km. As dispersal ability varies widely among EcM species (Ashkannejhad & Horton 2006) variation in dispersal of fungal propagules may have delayed the reestablishment of certain EcM fungi at particular sites and warrants further investigation. In addition to the large amount of unexplained variation in fungal community composition, several other limitations are present in our study. First, although our molecular methods used to sequence DNA were kept relatively constant, there were minor deviations in protocols. Differences in DNA extraction methods, DNA polymerase, primer sets, and PCR

conditions can result in major changes to estimates of microbial diversity (Brandariz-Fontes et al. 2015; Fouhy et al. 2016). Here, the DNA extraction method was kept the same across years with similar DNA polymerase and cycling conditions. Although there were differences in primer sets across years, (1) Sanger sequencing was performed across datasets and (2) all primers across datasets apply to the ITS region of fungal nuclear rDNA (Martin & Rygielwicz 2005). Moreover, in silico PCR analysis indicated that the coverage of ITS primer sets used was high and the taxonomic bias of amplification is expected to be low without a pronounced effect on downstream analyses of fungal community structure. Second, our data analysis methods used multiple datasets with differences in sampling depth and representative sequences. Differences in these approaches are also known to have a significant impact on the interpretation of microbial community structure (Golob et al. 2017; Xue et al. 2018). We accounted for these differences in our bioinformatics approach. Since primer sets were different across datasets, (1) we first trimmed paired end sequences for each dataset at the actual primer sites (see Martin & Rygielwicz 2005 for primer locations and positions), (2) performed quality filtering and clustering separately for each dataset, and (3) then merged the datasets together. This, a similar method to a model-based approach, currently implemented to correct amplicon errors in high-throughput sequencing (Callahan et al. 2016).

Taken together, our results indicated that common management interventions cannot alone reestablish EcM fungal communities in these novel ecosystems in the short term (i.e. “Carbon Copy Myth” sensu; Hilderbrand et al. 2005). First, it should be recognized that these EcM fungal communities will continue to change in both species composition and relative abundance across the maturing reclaimed areas. Longer-term monitoring should be incorporated to better understand and track how differences in host and cover soils interact to affect below-ground EcM fungal communities. Understanding how time, host-mediated effects, cover soil type, and disturbance severity influence the composition of EcM fungi is of broad ecological significance as these organisms are critical for the functioning of early and secondary successional systems with important implications for seedling establishment and ecosystem restoration.

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Supporting Information

The following information may be found in the online version of this article:

Figure S1. Layout of the reclamation site, showing reclaimed soil treatments of peat, forest floor material, and subsoil.

Figure S2. Layout of plots at the reference site used to compare ectomycorrhizal fungal communities across a severity of disturbance treatments with those assayed at a 4-year-old reclamation site.

Figure S3. Coverage of fungal ITS primers used in the molecular identification of ectomycorrhizal fungi assayed from tree seedlings on both reclaimed and reference sites in the Central Mixedwood subregion of the Canadian boreal forest.

Appendix S1. Molecular identification of ectomycorrhizal fungi assayed from tree seedlings on both reclaimed and reference sites in the Central Mixedwood subregion of the Canadian boreal forest.

Table S1. Primers used in the current study.

Table S2. Ectomycorrhizal fungi assayed from seedlings on a 4-year-old reclamation site within the Aurora North mine on the Syncrude Canada Ltd.

Table S3. Results of permutational multivariate analysis of variance testing the effects of time, host identity and reclaimed cover soils and their interactions on the community composition of root-associated ectomycorrhizal fungi in northern Alberta, Canada.

Table S4. Results of permutational multivariate analysis of variance testing the effects of disturbance severity, host identity, and their interactions on root-associated ectomycorrhizal fungal community composition of 4-year-old planted seedlings in northern Alberta, Canada.

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